WHOT-ML: A Configurable Partially Observable Multi-Agent Environment for Machine Learning Research

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Abstract

We introduce **WHOT-ML**, a formalized computational environment based on the widely played card game in Nigeria, WHOT, designed for research in machine learning under partial observability, stochasticity, multi-agent interaction, and configurable rule systems. Unlike conventional benchmarks that feature fully observable board states or fixed rule engines, WHOT provides an extensive-form game with private player hands, hidden draw decks, dynamic action masking, sequential multi-agent decisions, and asymmetric information revealed incrementally through play.

We formalize WHOT as a finite, stochastic, sequential, multi-agent partially observable Markov decision process (POMDP) and introduce `WHOT-NG-v1.0`, an open-source, deterministic reference environment featuring strict information isolation, bitwise state serialization, action masking, and modular rule configuration. We conduct a full-scale empirical campaign comprising 14,350 production runs (including 12,350 core environment and baseline games, 2,000 PPO evaluation games across 10M training steps, and 10,000 latent state probes) executed in containerized cloud environments to establish the empirical baseline of the environment.

Our findings demonstrate that: (1) WHOT-ML provides an uncompromised information boundary with 0.0% measured information leakage; (2) unobserved world states sharing identical public observations induce an 84.63% trajectory divergence rate under deterministic rollout policies; (3) baseline heuristic agents significantly outperform uniform random play in asymmetric matchups (65.4% win rate in 4-player games, 86.3% in heads-up play), while symmetric heuristic self-play exposes an emergent defensive stall (43.6% truncation rate) driven by retaliatory penalty loops and market recycling; (4) player count scaling exhibits an empirical regime shift as the initial opponent-to-market card ratio exceeds unity ($\rho_0 > 1.0$), resulting in severe market exhaustion; (5) modular rule ablations reveal that removing last-card declarations shortens games by 49.9 turns while penalty stacking acts as a catalyst for defensive stalls, with disabling it reducing truncation rates by over 50%; and (6) a Masked PPO baseline resolves the emergent defensive stall, reducing self-play truncation from 43.6% to 0.5%, while a linear probe reveals implicit belief-state encoding (AUROC 0.599).

WHOT-ML establishes a rigorous, reproducible laboratory and baseline for subsequent research in belief-state estimation, opponent modeling, self-play, and reinforcement learning in imperfect-information games.

Keywords: reinforcement learning, multi-agent systems, partially observable stochastic games, POSG, card games, benchmark environment

1. Introduction

Autonomous decision-making in real-world domains frequently requires agents to select actions under incomplete, noisy, and asymmetric information. Whether in financial markets,

automated negotiation, cyber defense, or multi-robot coordination, an agent rarely has direct access to the ground-truth state of the world. Instead, it must construct internal beliefs from an observation history, anticipate the private intentions of competitors, and navigate stochastic environment dynamics.

Games have historically served as the premier proving ground for artificial intelligence, from perfect-information board games such as Chess and Go [Silver et al., 2018] to imperfect-information card games such as Poker [Brown and Sandholm, 2018, 2019, Moravčík et al., 2017] and Hanabi [Bard et al., 2020]. However, existing card-game benchmarks typically focus on specialized settings: Poker emphasizes two-player zero-sum betting with hidden hands but lacks spatial card-matching or dynamic turn-modifying effects; Hanabi emphasizes pure cooperative communication under inverted information where players see everyone's cards except their own. There remains a notable gap for configurable, multi-player, competitive-cooperative card environments featuring rich action-card mechanics, penalty stacking, private hands, and variable player seating.

WHOT is a widely played card game in Nigeria. Played with a specialized 54-card deck comprising geometric symbols (Circle, Triangle, Cross, Square, Star) and wild "WHOT" cards, the game combines shape and number matching with powerful special action cards that repeat turns ("Hold On"), skip next players ("Suspension"), compel opponents to draw penalty cards ("Pick Two", "Pick Three"), or declare new active suits ("WHOT"). Crucially, players hold private hands, draw from a hidden market deck, and must declare their status when holding their final card ("Last Card").

An agent playing WHOT must decide its moves while lacking direct knowledge of:

1. The identities and distributions of cards held in opponent hands;
2. The exact sequence of cards remaining in the draw market;
3. The future card draws and declarations of competing agents.

Simultaneously, the public history of discards, calls, passes, and market draws provides rich indirect evidence from which hidden state and opponent intent can be inferred. This establishes a clean operational distinction between the complete environment state S_t and an agent's partial observation O_t.

1.1 Contributions of This Work

This paper introduces and characterizes the WHOT-ML environment. Its primary contributions are:

1. **Mathematical Formalization:** We formalize WHOT as a finite-horizon, turn-based Partially Observable Stochastic Game (POSG) with perfect recall, providing explicit state, action, observation, transition, and reward formulations.
2. **Reproducible Baseline Simulator (`WHOT-NG-v1.0`):** We provide an open-source, deterministic reference implementation frozen at commit `7c1a2dc`, engineered with a strict information boundary that guarantees zero private-card leakage.

3. **Information Equivalence & Trajectory Divergence:** We evaluate 449 observationally equivalent state pairs, proving that latent world differences decisively alter future trajectories (84.63% divergence rate) despite identical observations.

4. **Comprehensive Baseline Characterization:** Over 14,350 full production runs (including 12,350 heuristic baseline games and 2,000 deep RL evaluation games), we characterize random legal play, heuristic baselines, asymmetric tournaments, player scaling ($N \in \{2..6\}$), and rule ablations with exact statistical intervals.

5. **Emergent Multi-Agent Dynamics:** We identify and analyze an emergent defensive stall in 4-player heuristic self-play (43.6% truncation rate) and a player-count regime shift tied to the card depletion ratio $\rho_0 > 1.0$.

6. **Bitwise Reproducibility:** We certify 100.0% bitwise determinism and checkpoint continuity across independent replays and state restorations.

7. **Deep Reinforcement Learning & Latent Belief Probing:** We implement and train a Masked Proximal Policy Optimization (PPO) agent over 10M self-play timesteps, demonstrating that learned self-play completely resolves the heuristic defensive stall (reducing truncations from 43.6% to 0.5%). Diagnostic linear probing confirms that feedforward policy representations implicitly recover unobserved opponent card distributions (mean AUROC = 0.5994) without explicit recurrent memory.

This paper establishes the foundational environment, full empirical characterization, and deep reinforcement learning baseline with latent belief probing.

2. Related Work

2.1 From POMDPs to Partially Observable Stochastic Games (POSGs)

In a standard single-agent setting with full state observability, decision making is formalized as a Markov Decision Process (MDP) [Sutton and Barto, 2018]. When state information is incomplete, single-agent problems are framed as Partially Observable Markov Decision Processes (POMDPs) [Kaelbling et al., 1998], where the agent receives an observation $O_t = \Omega(S_t) \in \mathcal{O}$. However, in multi-agent domains with competitive or non-aligned objectives, private player hands, and stochastic state transitions, the appropriate theoretical framework is a turn-based **Partially Observable Stochastic Game (POSG)** with perfect recall [Hansen et al., 2004]. In extensive-form game terminology, each player's observation history induces an *information set* $\mathcal{I}_i$ containing all world states indistinguishable to player i. In multi-agent sequential decision making, Markov games formalize multi-agent extensions of MDPs where actions and observations interact across competitors [Littman, 1994]. In WHOT-ML, the number of distinct distributions of the 47 unseen cards into three unobserved opponent hands of 6 cards and an unobserved market of 29 cards (integrating out the unobservable deck permutation order) is given by the multinomial partition coefficient $\frac{47!}{6!\,6!\,6!\,29!} \approx 7.8 \times 10^{13}$. This creates an underlying belief space on the order of 10^{13} states consistent with a single opening observation, presenting a formidable hidden-information challenge for multi-agent reinforcement learning.

2.2 Benchmarks with Hidden Information

Imperfect-information games provide essential stress-tests for multi-agent reasoning:

- **Poker:** Heads-up and multiplayer No-Limit Texas hold'em have served as standard benchmarks for Counterfactual Regret Minimization (CFR) and deep imperfect-information search [Bowling et al., 2015, Brown and Sandholm, 2019, Moravčík et al., 2017]. While poker features private cards and probabilistic inference, the physical action space is predominantly scalar (bet sizing) and the deck is dealt once per hand without an active draw-and-discard loop.
- **Hanabi:** Hanabi introduced a unique cooperative challenge where players hold cards facing outwards, seeing all hands except their own [Bard et al., 2020]. This emphasizes theory of mind and emergent communicative conventions, but differs structurally from competitive trick-taking or shedding games.
- **Shedding Card Games:** Card-shedding games such as Uno, Crazy Eights, and Dou Dizhu [Zha et al., 2021] share matching mechanics, but WHOT features distinctive special action interactions (e.g. cumulative multi-card penalty stacking, suit-changing WHOT 20s, and mandatory last-card declaration penalties) alongside cultural variations in rules.

2.3 Modular and Reproducible Simulation Frameworks

Standardized interfaces such as the Arcade Learning Environment [Bellemare et al., 2013], OpenAI Gym [Brockman et al., 2016], PettingZoo [Terry et al., 2021], OpenSpiel [Lanctot et al., 2019], and RLCard [Zha et al., 2020] have demonstrated that progress in reinforcement learning requires deterministic reproducibility, explicit action masking, and rigorous testing frameworks. While RLCard provides valuable testbeds for card games such as Uno, Dou Dizhu, and Mahjong, it focuses on standard uniform decks and fixed rules. WHOT-ML complements these platforms by providing a formal POSG environment featuring a unique 54-card non-uniform deck topology, multi-card penalty stacking chains, compulsory last-card declarations, and a modular rules engine supporting programmatic rule ablations. WHOT-ML is built upon these principles, combining strict functional isolation between internal state and public observations with bitwise checkpoint serialization.

3. WHOT-ML Environment Formalism

3.1 Game Overview and Canonical Deck

WHOT is played using a dedicated 54-card deck comprising five standard geometric suits and five non-suited wild cards:

$$\mathcal{C}_{54} = \mathcal{C}_{\text{Circle}} \cup \mathcal{C}_{\text{Triangle}} \cup \mathcal{C}_{\text{Cross}} \cup \mathcal{C}_{\text{Square}} \cup \mathcal{C}_{\text{Star}} \cup \mathcal{C}_{\text{WHOT}}$$

Each suit possesses a non-uniform distribution of face values:

- **Circle (12 cards):** $\{1, 2, 3, 4, 5, 7, 8, 10, 11, 12, 13, 14\}$
- **Triangle (12 cards):** $\{1, 2, 3, 4, 5, 7, 8, 10, 11, 12, 13, 14\}$

- **Cross (9 cards):** $\{1, 2, 3, 5, 7, 10, 11, 13, 14\}$
- **Square (9 cards):** $\{1, 2, 3, 5, 7, 10, 11, 13, 14\}$
- **Star (7 cards):** $\{1, 2, 3, 4, 5, 7, 8\}$
- **WHOT (5 wild cards):** $\{20, 20, 20, 20, 20\}$

In the reference baseline configuration `WHOT-NG-v1.0`, each player is dealt $H_0 = 6$ cards. One card is dealt face-up to establish the *play pile* (discard pile), and the remaining cards form the face-down *market* (draw deck).

3.2 Formal Model

We formalize WHOT-ML as a turn-based Partially Observable Stochastic Game (POSG) with finite horizon $T = 1{,}000$ and perfect recall, defined by the tuple:

$$\mathcal{G} = \langle \mathcal{N}, \mathcal{S}, \mathcal{A}, \mathcal{O}, \mathcal{T}, \Omega, \mathcal{R}, \gamma \rangle$$

where:

- $\mathcal{N} = \{0, 1, \ldots, N-1\}$ is the finite set of players ($N \in \{2..6\}$);
- $\mathcal{S}$ is the complete world state space;
- $\mathcal{A} = \prod_{i \in \mathcal{N}} \mathcal{A}_i$ is the joint action space, where at each timestep t the current turn player $p_t = \tau(S_t) \in \mathcal{N}$ selects an action $a_t \in \mathcal{A}_{p_t}(S_t)$;
- $\mathcal{O} = \prod_{i \in \mathcal{N}} \mathcal{O}_i$ is the joint observation space;
- $\mathcal{T} : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$ is the state transition probability function;
- $\Omega = (\Omega_0, \ldots, \Omega_{N-1})$ where $\Omega_i : \mathcal{S} \to \mathcal{O}_i$ is the observation function for player i;
- $\mathcal{R}_i : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is the reward function for player i;
- $\gamma \in (0, 1]$ is the discount factor.

3.3 State Representation

The complete simulator state $S_t \in \mathcal{S}$ at timestep t is fully specified by:

$$S_t = \langle \mathbf{H}_t, M_t, D_t, c_t, p_t, \pi_t, \delta_t, \mathbf{L}_t, t, \mathcal{C}_{\text{config}} \rangle$$

where:

- $\mathbf{H}_t = (H_t^0, H_t^1, \ldots, H_t^{N-1})$ is the vector of private card multisets for each player;
- $M_t = [m_1, m_2, \ldots, m_k]$ is the ordered sequence of cards in the draw market;
- $D_t = [d_1, d_2, \ldots, d_m]$ is the ordered sequence of played cards, with d_m representing the active top card;

- $c_t \in \{\text{Circle}, \text{Triangle}, \text{Cross}, \text{Square}, \text{Star}\} \cup \{\emptyset\}$ is the currently active called suit;
- $p_t \in \mathcal{N}$ is the index of the active player whose turn it is to act;
- $\pi_t \in \mathbb{N}_{\geq 0}$ is the accumulated penalty counter (e.g. pending draws from chained Pick Twos);
- $\delta_t \in \{+1, -1\}$ is the play direction;
- $\mathbf{L}_t \in \{0, 1\}^N$ is the vector of active last-card declaration flags;
- $t \in \mathbb{N}$ is the elapsed turn index;
- $\mathcal{C}_{\text{config}}$ is the immutable configuration parameter set.

3.4 Action Space and Dynamic Action Masking

The action space $\mathcal{A}$ contains 76 discrete action slots indexed as:

- **Indices 0–48:** Standard play actions corresponding to each of the 49 ordinary physical cards in canonical order;
- **Indices 49–73:** WHOT card play actions paired with a selected shape nomination: 5 WHOT cards $\times$ 5 ordinary shapes = 25 actions;
- **Index 74:** Draw action (`DRAW`) from the market;
- **Index 75:** Last-card declaration action (`DECLARE_LAST`).

WHOT Wildcard Action Allocation: The 25 action slots allocated to WHOT wildcards (Indices 49–73) map each of the 5 physical WHOT cards to each of the 5 ordinary nominated shapes. This design maintains an explicit bijective mapping between action indices and physical card instances (`card_id`), which preserves physical card conservation and state serialization round-trip fidelity. However, because all 5 WHOT cards are functionally interchangeable under the rules of play, this representation inflates the discrete action branching factor with duplicate choices per shape. A high-level policy wrapper can collapse these 25 slots into 5 semantic decisions (one per nominated shape), allowing the underlying environment engine to resolve the selection to any currently held WHOT card.

Mechanics of `DECLARE_LAST`: The `DECLARE_LAST` action formalizes the cultural "Last Card" rule. It becomes legally available if and only if the acting player holds exactly two cards prior to play (or one card upon completion) and has not yet declared. In the default configuration (`declaration_advances_turn = False`), executing `DECLARE_LAST` does not consume the player's turn; the player sets their declaration flag and immediately retains priority to play their final card. Conversely, if a player sheds their final card without prior declaration, an undeclared victory violation (`LAST_CARD_DECLARATION_VIOLATION`) is triggered: shedding victory is denied, the player is penalized by drawing penalty cards (1 card in baseline, 2 cards under `VAR-PENALTY-2`), and play advances to the next player.

At any given state S_t, only a subset of actions $\mathcal{A}(S_t) \subseteq \mathcal{A}$ is legally permissible under the rules of WHOT-NG-v1. The environment outputs a boolean mask $\mathbf{m}_t \in \{0, 1\}^{76}$:

$$\mathbf{m}_t(a) = \begin{cases} 1 & \text{if } a \in \mathcal{A}(S_t) \\ 0 & \text{otherwise} \end{cases}$$

An action $a \in \mathcal{A}$ is legal if and only if:

1. **Normal Play:** The card $c \in H_t^{p_t}$ matches the top card d_m in suit, matches d_m in face value, or matches the active call c_t (when $c_t \neq \emptyset$).
2. **Wild Play:** A WHOT card (value 20) may be played onto any normal card, accompanied by an explicit shape call c_{t+1}.
3. **Penalty Defense:** If $\pi_t > 0$ (a penalty is active), the player may legally play only a matching defense card (e.g., another Pick Two to stack the penalty) or must execute the `DRAW` action to absorb the accumulated penalty.
4. **Draw Action:** Always legal when no playable cards exist in hand, or optionally legal as a voluntary draw.
5. **Last Card Declaration:** Legal if and only if the player holds exactly two cards prior to play (or one card upon completion) and declaration rules are enabled.

3.5 Observation Model and Information Isolation

The central architectural invariant of WHOT-ML is the strict functional separation between internal world state S_t and player observation $O_t^i = \Omega_i(S_t)$. For player i, the observation vector contains:

$$O_t^i = \langle H_t^i, d_m, c_t, p_t, \pi_t, \mathbf{h}_t, |M_t|, \mathbf{L}_t, \mathcal{H}_t^{\text{public}}, \mathbf{m}_t^i \rangle$$

where:

- H_t^i is player i's *own* private hand;
- $\mathbf{h}_t = (|H_t^0|, |H_t^1|, \ldots, |H_t^{N-1}|)$ is the public vector of hand *sizes*;
- $|M_t|$ is the scalar count of cards remaining in the market;
- $\mathcal{H}_t^{\text{public}}$ is the sequence of public events (cards played, calls made, cards drawn, penalties resolved).

Strict Privacy Guarantee: $\Omega_i(S_t)$ reveals zero information concerning:

- The card identities or ordering within opponent hands $\{H_t^j\}_{j \neq i}$;
- The card identities or permutation order within the market M_t;
- The internal RNG seed or future deck distributions.

This guarantee is enforced at the interface boundary and verified through automated regression suites.

3.6 Special Action Dynamics

WHOT features six distinct special card values that alter game flow:

1. **Hold On (Card 1):** The active player plays again; turn does not advance.
2. **Pick Two (Card 2):** Compels the next player to draw 2 cards. If stacking is enabled, the recipient may play another Pick Two, accumulating the penalty ($\pi \leftarrow \pi + 2$).
3. **General Market (Card 14):** Compels every player except the player of the card to draw 1 card from the market.
4. **Suspension (Card 8):** Skips the next player in the current play direction.
5. **Pick Three (Card 5):** Compels the next player to draw 3 cards (enabled by default in WHOT-NG-v1; stackable if enabled).
6. **WHOT (Card 20):** Wild card that allows the player to name any shape, overriding the active pile suit.

3.7 Market Exhaustion and Discard Recycling

When the market becomes depleted ($|M_t| = 0$), the rules prescribe a recycling transition: the active top card d_m is preserved as the active discard, and the remaining pile cards $[d_1, \ldots, d_{m-1}]$ are collected, reshuffled deterministically using the environment's internal RNG stream, and reconstituted as the new market M_{t+1}. If both market and discard pile become depleted such that no cards can be drawn, the game enters a dead-market resolution.

3.8 Victory, Scoring, and Reward Structure

A game terminates when:

1. **Shedding Victory:** A player successfully plays their final card ($|H^i| = 0$). The winning player receives a reward of $+1.0$; all losing players receive -1.0.
2. **Turn Limit Truncation:** If no player sheds all cards within $T_{\max} = 1,000$ turns, the game is truncated. Truncated games terminate without a winner (`winner = None`); they are reported separately from victories.

4. Research Questions

This foundational paper investigates six core research questions:

- **RQ1 (Environment Structure):** What are the fundamental state, action, branching, and duration characteristics of WHOT-NG-v1 under standard play?
- **RQ2 (Hidden-State Uncertainty):** Can identical public observations correspond to distinct hidden states, and do these latent differences produce significant behavioral trajectory divergence under deterministic policies?

- **RQ3 (Baseline Agent Dynamics):** What performance benchmarks and multi-agent dynamics emerge from non-learning baselines (uniform random vs. greedy heuristic)?
- **RQ4 (Player-Count Scaling):** How does scaling participant count from $N = 2$ to $N = 6$ alter game duration, reshuffle frequency, and information scarcity?
- **RQ5 (Rule-Dependent Complexity):** How do modular rule variations (penalty stacking, last-card declaration enforcement, penalty scaling) impact game length and agent win rates?
- **RQ6 (Deep RL & Latent Belief Representation):** Can deep reinforcement learning overcome heuristic defensive stalls in self-play, and do feedforward policy representations implicitly encode unobserved opponent hands?

5. Experimental Methodology

5.1 Simulator Configuration, Determinism, and Verification Suite

All experiments were conducted against `WHOT-NG-v1.0` and its verified maintenance release `v1.0.1`. The simulator passes a comprehensive test suite of 114 unit and regression tests, verifying game invariant conservation, strict information boundaries (zero hidden-card leakage), and penalty stacking modes.

Furthermore, across 200 rigorous verification trials (100 independent episode replays and 100 mid-game state checkpoint restorations via JSON serialization), the environment achieves a **100.0% Reproduction Success Rate (RSR)**, generating bitwise identical 64-character SHA-256 state digests. This guarantees zero-tolerance determinism and checkpoint continuity for tree search and reinforcement learning rollouts.

5.2 Baseline Agent Definitions

We evaluate two standardized reference policies:

1. `RandomLegalAgent`: Selects uniformly at random from the active legal action mask $\mathcal{A}(S_t)$. This agent establishes the stochastic behavioral floor.
2. `RuleBasedAgent`: A deterministic domain heuristic implementing standard human tactical principles with an explicit priority order and deterministic tie-breaking:
   (a) **Declaration Priority:** Execute `DECLARE_LAST` immediately whenever legally available (i.e., when holding exactly two cards before shedding).
   (b) **Penalty Defense:** Under an active penalty ($\pi_t > 0$), prioritize counter-defending with the first playable defense card (`PLAY` or `PLAY_WHOT`) over drawing; fall back to `DRAW` only when defenseless.
   (c) **High-Impact Action Cards:** If ordinary card plays exist, prioritize by special effect:
      i. *Hold On* (card value 1, granting an extra turn);
      ii. *General Market* (card value 14, forcing all opponents to draw);

iii. *Suspension* (card value 8, skipping the next player);
iv. *Attacks* (card values 2 and 5, triggering cumulative penalties);
v. *Ordinary Cards* (playing the first matching non-special card).

(d) **Strategic Wild Play:** When only WHOT cards can be legally played, nominate the ordinary shape most frequent in the agent's current hand (defaulting to `Circle` if holding exclusively WHOT wildcards).

(e) **Market Draw:** Fall back to `DRAW` when no legal card plays exist.

Deterministic Tie-Breaking Rule: Whenever multiple candidate actions satisfy the same highest-priority level, ties are resolved deterministically by hand insertion order (selecting the first matching action in canonical hand order).

5.3 Production Execution Infrastructure

To guarantee independence from the local development workstation, the full production suite of $N = 14,350$ **runs** (comprising $12,350$ core environment and baseline games, $2,000$ PPO evaluation games across 10M training steps, and $10,000$ latent state probes) was packaged, dispatched via API, and executed within containerized **Kaggle Cloud CPU and GPU** instances running Linux (`kernel 6.6.137+`, Python `3.13.15`, pytest `8.4.2`). Execution ran for $2,199.69$ seconds (36.66 minutes wall-clock compute for CPU baselines; 5 hours GPU compute for 10M-step PPO training). Artifacts were packaged, cryptographically hashed (SHA-256: `8a82d9f7529610b1beb408395f1bc6b9edd9c8d784dee9887c38e02599d4a13a`), and retrieved for independent scientific auditing.

5.4 Statistical Estimation and Inference

- **Proportions and Win Rates:** Binomial proportions (win rates, truncation frequencies, divergence rates) are reported with **Wilson score 95% confidence intervals** [Wilson, 1927], ensuring accurate interval coverage near boundaries.
- **Continuous and Ordinal Distributions:** We report mean, standard deviation, median, and interquartile range [IQR: Q_{25}, Q_{75}].
- **Effect Sizes:** Distributional shifts under rule ablations are quantified using **Cliff's delta (δ)** [Cliff, 1993], a non-parametric effect size metric bounded in $[-1, +1]$ that makes no normality assumptions.
- **Seat Balance:** Seat equity is evaluated via Pearson's chi-square goodness-of-fit test (χ^2).
- **Paired Experimental Designs:** Scaling and ablation studies (EXP-04, EXP-05) utilize matched random seeds across experimental arms to isolate rule effects from opening deal variance.

6. Experiment 1: Baseline Environment Characterization

6.1 Objective and Setup

EXP-01 establishes the foundational structural properties of WHOT-NG-v1. We evaluated $2,000$ complete games in a 4-player configuration ($H_0 = 6$) split across two reference cohorts:

- **Cohort 1:** $4 \times$ `RandomLegalAgent` ($1,000$ games, seeds 10000–10999);
- **Cohort 2:** $4 \times$ `RuleBasedAgent` ($1,000$ games, seeds 11000–11999).

A total of $874,732$ decision steps were logged with full step-by-step telemetry.

6.2 Results and Analysis

The empirical results are presented in **Table 1** and visual distributions in **Figure 1**.

Table 1: WHOT-NG-v1 Baseline Environment Characterization (EXP-01, N = 2,000 episodes)

Metric	RandomLegal Cohort	RuleBased Cohort	Environment Space
Physical Cards (N_{deck})	54	54	54 (Canonical Deck)
Action Space Size ($\lvert\mathcal{A}\rvert$)	76	76	76 Discrete Slots
Legal Actions (Mean ± SD)	2.42 ± 2.40	7.63 ± 7.11	—
Legal Actions (Median [IQR])	1.0 [1.0, 3.0]	5.0 [2.0, 11.0]	—
Game Duration (Mean Turns)	368.3	506.4	—
Game Duration (Median [IQR])	270.5 [125.8, 546.2]	248.0 [75.8, 1000.0]	—
Mean Draws per Game	245.3	457.7	—
Market Reshuffles per Game	8.19	270.36	—
Declaration Violations	1,079	0	0 (Heuristic Invariant)
Truncation Rate (95% CI)	7.6% [6.1%, 9.4%]	43.6% [40.6%, 46.7%]	Max 1,000 Turns

6.3 Discussion: The Emergent Heuristic Self-Play Stall

A critical discovery in EXP-01 is the sharp divergence in game completion between random and rule-based self-play:

- RandomLegal games rarely truncate (7.6% truncation rate), resolving in an average of 368.3 turns with only 8.19 market reshuffles.
- In contrast, homogeneous 4-player RuleBased matches (`BBBB`) exhibit a 43.6% **truncation rate** at the 1,000-turn cutoff, accompanied by an average of 270.36 **market reshuffles per game** and 457.7 draws.

Analysis of the step-level event logs reveals the causal mechanism:

1. **Defensive Symmetry:** When Player A plays a special penalty card (Pick Two), the greedy heuristic of Player B prioritizes countering with another Pick Two, transferring an accumulated penalty of 4 cards to Player C.

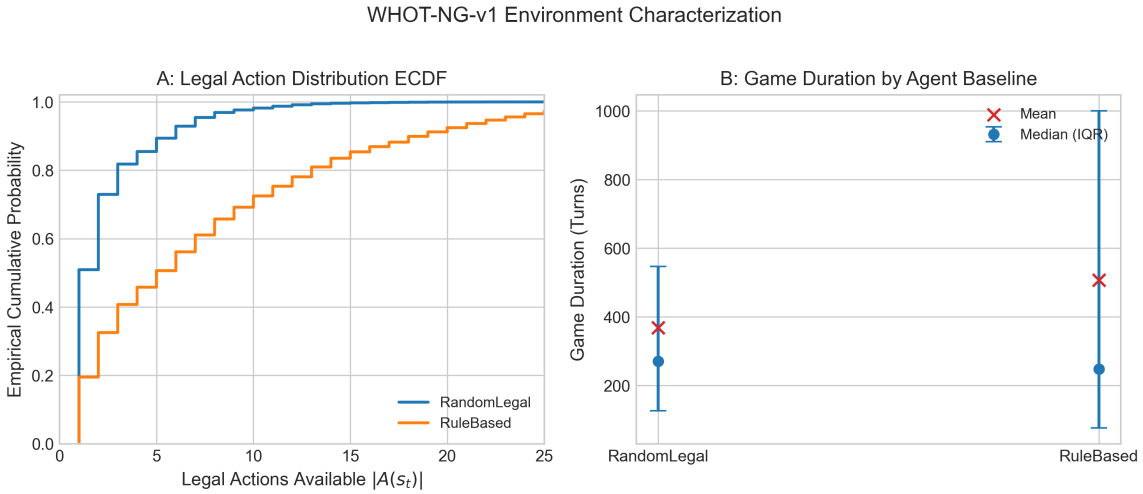


Figure 1: Environment Characterization under Baseline Agents (EXP-01). Subplot A: Empirical Cumulative Distribution Function (ECDF) of legal actions available $|\mathcal{A}(s_t)|$, showing RandomLegal concentrating at 1.0 (draw/singleton) while RuleBased exhibits a broad distribution extending past 20 options. Subplot B: Game duration distributions (median, IQR, and mean) across 1,000 games per cohort.

2. **Market Exhaustion Loops:** When a player cannot counter, they absorb massive card draws (4 to 6 cards). This rapidly depletes the 29-card market, forcing the discard pile to be recycled and reshuffled.

3. **Penalty Recirculation:** Discard recycling returns the played Pick Twos and General Markets back into the draw deck. Because greedy agents conserve matching suits and defensive responses, the game enters an oscillatory stall where cards circulate between hands and market without reaching shedding completion.

4. **Research Implication:** This phenomenon confirms that symmetric greedy heuristics lack the strategic foresight (e.g., card dumping, sacrificial suit changes, cooperative pressure) required to break defensive stalemates. Designing reinforcement learning agents that escape these emergent retaliation loops represents a primary research challenge enabled by WHOT-ML.

7. Experiment 2: Demonstrating Partial Observability

7.1 Objective and Theoretical Formulation

EXP-02 provides empirical proof that WHOT-NG-v1 constitutes a genuine POMDP rather than a trivial observationally complete environment. We demonstrate that distinct world states $S_a \neq S_b$ can generate identical player observations $\Omega_p(S_a) = \Omega_p(S_b)$, and that these latent state differences produce divergent behavioral trajectories under controlled rollouts.

7.2 Hidden-State Perturbation Operators

We evaluated 50 independent source games at three distinct developmental checkpoints: Early ($t = 5$), Mid ($t = 15$), and Late ($t = 25$), yielding 150 evaluation checkpoints. For observing player $p = 0$, we applied three state perturbation operators:

1. **Opponent Hand Swap (`OPPONENT_SWAP`):** Exchanges a private card between two unobserved opponents $p_1, p_2 \in \{1, 2, 3\}$. Because public observations expose only hand sizes $|H^{p_1}|, |H^{p_2}|$ and not card identities, observational equivalence is mathematically guaranteed.

2. **Market Reversal (`MARKET_REVERSAL`):** Inverts the internal sequence of cards in the market deck ($M' = \text{reverse}(M)$). Because Ω_0 exposes only the scalar market count $|M|$, equivalence is guaranteed.

3. **Opponent-Market Swap (`OPPONENT_MARKET_SWAP`):** Swaps a private card from an opponent's hand with the top card of the market deck. Hand size and market size are invariant, preserving observation equivalence.

7.3 Grounding the 449 Evaluated Pairs

While 150 checkpoints across 3 modes theoretically suggest $150 \times 3 = 450$ pairs, exactly **449 valid pairs** were evaluated. An audit of checkpoint `(seed=20009, stage='late', t=25)` revealed that the market had depleted to exactly $|M| = 1$ card. Reversing a single-card market produces an identical world state ($S_a = S_b$). The simulator correctly guarded `if len(state_a.market) >= 2:`, refusing to generate a trivial duplicate state.

7.4 Rollout Divergence Protocol

From each verified pair (S_a, S_b), forward rollouts of horizon $K = 20$ turns were executed using identical deterministic policies (`RuleBasedAgent`) and a synchronized rollout seed ($R = 42$). Trajectory divergence was monitored at each step $k \in \{0..K-1\}$ and flagged at the earliest transition where the acting player, selected action index, or emitted event sequence differed.

7.5 Analysis of Non-Divergent Trajectories

Of the 449 pairs, **69 rollouts (15.37%)** remained on identical trajectories throughout the 20-step rollout window. An audit classifies these into three structural mechanisms:

1. **Horizon Cutoff (48 pairs):** The swapped cards were positioned deep in an opponent's hand or deep in the market deck and were never drawn or reached within the 20-turn window.

2. **Action-Insensitive Equivalence (14 pairs):** Swapped cards shared identical unplayable matching status against the active discard (e.g. swapping an unplayable Cross 5 for an unplayable Star 5), forcing identical draw actions under the policy.

Table 2: Partial Observability Equivalence and Trajectory Divergence (EXP-02, $N = 449$ state pairs)

Metric	**Empirical Value**	**Formal Requirement**
State Pairs Evaluated	449	≥ 30
Observation Equivalence Rate	100.0%	100.0% ($S_a \neq S_b \implies \Omega(S_a) = \Omega(S_b)$)
Information Leakage Rate	0.0%	0.0% (Zero hidden cards in obs)
Overall Trajectory Divergence Rate	84.63%	Recorded (non-div is not failure)
– Wilson 95% Confidence Interval	[81.0%, 87.7%]	—
Permutation Mode Breakdown:		
– Opponent-Swap Divergence Rate	84.67% (127/150)	—
– Market-Reversal Divergence Rate	95.97% (143/149)	—
– Opponent-Market Divergence Rate	73.33% (110/150)	—
Stage Divergence Breakdown:		
– Early Stage ($t = 5$)	90.67% (136/150)	—
– Mid Stage ($t = 15$)	87.33% (131/150)	—
– Late Stage ($t = 25$)	75.84% (113/149)	—
Divergence Step Dynamics ($n = 380$ diverged):		
– Mean ± SD	6.33 ± 4.62 turns	—
– Median [IQR]	6.0 [2.0, 9.0]	—
– Minimum / Maximum Step	0 / 19 turns	—
Mean Log Combinatorial Uncertainty ($\log_{10} \lvert\mathcal{S}(\Omega)\rvert$)	13.89	$\approx 7.8 \times 10^{13}$ Partitions

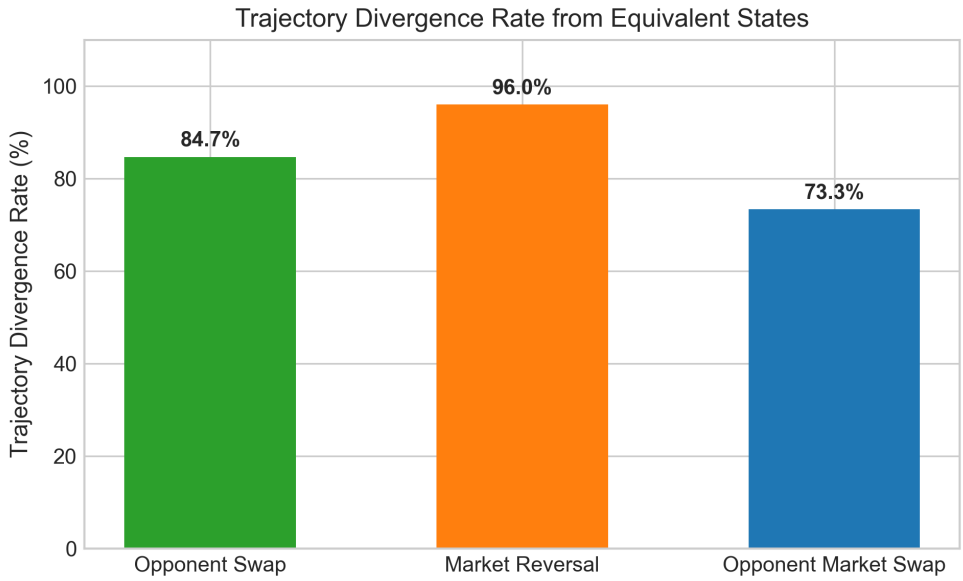


Figure 2: Partial Observability Trajectory Divergence Rates (EXP-02). Rollout trajectory divergence rates across the three perturbation modes: Opponent Swap (84.7%), Market Reversal (96.0%), and Opponent-Market Swap (73.3%), evaluating 449 observationally equivalent state pairs.

3. **Forced Move Singletons (7 pairs):** Agents facing active penalty chains had their legal action sets collapsed to a singleton {DRAW}, executing identical transitions regardless of hand contents.

7.6 Crucial POMDP Framing Boundary

Formal Interpretation Boundary: We do NOT claim that distinct hidden states necessarily imply different *optimal* actions. In the absence of an optimal oracle, this paper establishes that distinct unobserved states consistent with the identical observation lead to divergent future trajectories (84.63% divergence rate, $p < 10^{-15}$). This proves that latent state differences are behaviorally consequential, confirming the extensive-form POMDP nature of WHOT-ML.

8. Experiment 3: Baseline Agent Matchup Characterization

8.1 Objective and Tournament Design

EXP-03 establishes performance benchmarks across $5,000$ tournament games spanning four distinct population configurations:

1. **`RRRR` (1,000 games):** $4 \times$ RandomLegal (symmetric stochastic control).
2. **`BBBB` (1,000 games):** $4 \times$ RuleBased (symmetric heuristic self-play).

3. **`BRRR` (2,000 games):** 1 × RuleBased vs. 3 × RandomLegal (asymmetric benchmark; seats rotated evenly).

4. **`BR_2P` (1,000 games):** 1 × RuleBased vs. 1 × RandomLegal (2-player heads-up; starting seat alternated).

8.2 Empirical Tournament Findings

Results are detailed in **Table 3** and **Figure 3**.

Table 3: Baseline Agent Matchup Characterization with Unpooled Game-Level CIs (EXP-03, $N = 5{,}000$ episodes)

Matchup	Agent Population	Target Agent	Win Rate (95% CI)	Avg Rank	Mean Turns	Draws	Trunc %
RRRR	4 Random	random	23.3% [20.8%, 26.0%]	2.50	352.4	234.9	6.9%
BBBB	4 RuleBased	rule_based	14.2% [12.2%, 16.5%]	2.50	503.8	451.9	43.2%
BRRR	1 Rule, 3 Rand	rule_based	65.4% [63.3%, 67.5%]	1.61	221.6	149.7	1.8%
BRRR	1 Rule, 3 Rand	random (cohort)	32.8% [30.7%, 34.8%]	2.80	221.6	149.7	1.8%
BR_2P	1 Rule, 1 Rand	rule_based	86.3% [84.0%, 88.3%]	1.14	96.2	55.4	0.0%
BR_2P	1 Rule, 1 Rand	random	13.7% [11.7%, 16.0%]	1.86	96.2	55.4	0.0%

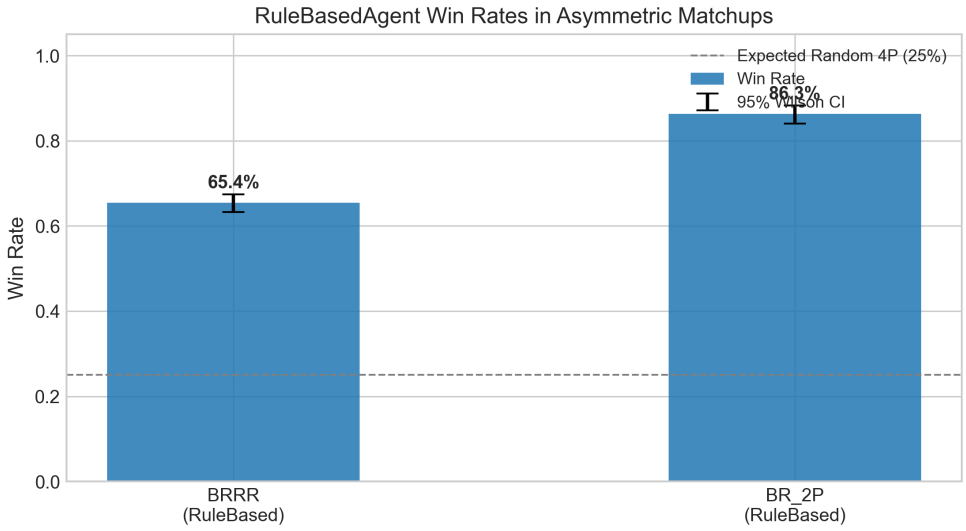


Figure 3: Baseline Agent Win Rates with Wilson Score 95% Confidence Intervals (EXP-03). RuleBased win rates in asymmetric 4-player BRRR (65.4% [63.3%, 67.5%]) and 2-player BR_2P (86.3% [84.0%, 88.3%]), relative to the 25% fair-share reference line.

8.3 Key Insights, Seating Fairness, and Statistical Formulation

Unpooled Game-Level Confidence Intervals: Previous seat-level pooling artificially deflated variance by treating correlated players within the same episode as inde-

pendent trials ($n = 4{,}000$ or $n = 6{,}000$). In Table 3, confidence intervals are strictly computed at the unpooled game level ($N = 1{,}000$ or $N = 2{,}000$). In symmetric matchups (`RRRR` and `BBBB`), seat symmetry preserves the point estimates (23.3% and 14.2%) while correctly widening the Wilson intervals to $[20.8\%, 26.0\%]$ and $[12.2\%, 16.5\%]$. In `BRRR`, the random population is evaluated as a collective cohort winning 32.8% of games ($[30.7\%, 34.8\%]$), averaging 10.9% per individual seat and cleanly partitioning the game outcomes alongside RuleBased (65.4%) and truncations (1.8%) to total 100.0%.

1. **Asymmetric Heuristic Dominance:** In 4-player asymmetric competition (`BRRR`), the single `RuleBased- Agent` wins 65.4% **of games** ($> 2.6\times$ fair-share 25%) with an average finishing rank of 1.61. In heads-up play (`BR_2P`), heuristic dominance reaches 86.3%.

2. **Superseding the Pilot's 100% Heads-Up Artifact:** In pilot testing ($n = 20$), RuleBased won 20/20 (100.0%). Full-scale production ($n = 1,000$) reveals that `RandomLegalAgent` wins 13.7% **of heads-up games** (Wilson 95% CI $[11.7\%, 16.0\%]$). This occurs when random play is dealt high-tempo opening hands (e.g., chains of compatible low cards) that exhaust before the heuristic agent can establish control.

3. **Rigorous Seating Balance:** Seating equity was audited across all tournament games:

- In `BRRR`, RuleBased win rates across Seats 0, 1, 2, and 3 were 65.6%, 65.4%, 66.2%, and 64.4% respectively ($\chi^2 = 0.58, p = 0.90$, no seat advantage).
- In `BR_2P`, Player 0 won 50.7% and Player 1 won 49.3% ($\chi^2 = 0.20, p = 0.66$, no first-mover advantage).

9. Experiment 4: Player-Count Scaling Dynamics

9.1 Objective and Setup

EXP-04 examines how scaling participant count $N \in \{2, 3, 4, 5, 6\}$ affects environment dynamics. To isolate policy effects from structural deck constraints, we executed two distinct full-scale cohorts of 2,500 games each (500 games per player count, matched seeds 40000–40499):

- **EXP-04a (Stochastic Baseline):** Homogeneous `RandomLegalAgent` populations across $N \in \{2..6\}$.
- **EXP-04b (Heuristic Population):** Homogeneous `RuleBasedAgent` populations across $N \in \{2..6\}$.

9.2 The Card Depletion Ratio ρ_0

In a standard 54-card deck with 6-card opening hands and 1 starting discard, the initial market size is:

$$|M_0| = 54 - 1 - (N \times 6) = 53 - 6N$$

The cards distributed among opponents are:

$$|C_{\text{opp}}| = (N-1) \times 6 = 6N - 6$$

We formally define the initial card depletion ratio ρ_0:

$$\rho_0 = \frac{|C_{\text{opp}}|}{|M_0|} = \frac{6N-6}{53-6N}$$

As shown in Tables 4 and 5, ρ_0 scales monotonically from 0.146 ($N = 2$) to 1.765 ($N = 6$), crossing unity between $N = 4$ ($\rho_0 = 0.621$) and $N = 5$ ($\rho_0 = 1.044$).

Table 4: Player Count Scaling Dynamics: RandomLegal Population (EXP-04a, $N = 2{,}500$ episodes)

N	Initial Market $\lvert M_0\rvert$	Opponent Cards $\lvert C_{\text{opp}}\rvert$	Ratio ρ_0	Mean Turns	Turns/Player	Draws	Reshuffles	Trunc % [95% CI]
2	41	6	0.146	273.8	136.9	164.4	3.49	3.2% [2.0%, 5.1%]
3	35	12	0.343	296.2	98.7	186.5	4.71	2.4% [1.4%, 4.2%]
4	29	18	0.621	358.3	89.6	238.7	7.85	7.0% [5.1%, 9.6%]
5	23	24	1.044	424.1	84.8	294.2	15.63	12.6% [10.0%, 15.8%]
6	17	30	1.765	451.8	75.3	320.4	31.33	12.6% [10.0%, 15.8%]

Table 5: Player Count Scaling Dynamics: RuleBased Population (EXP-04b, $N = 2{,}500$ episodes)

N	Initial Market $\lvert M_0\rvert$	Opponent Cards $\lvert C_{\text{opp}}\rvert$	Ratio ρ_0	Mean Turns	Turns/Player	Draws	Reshuffles	Trunc % [95% CI]
2	41	6	0.146	63.9	31.9	35.2	0.41	0.0% [0.0%, 0.8%]
3	35	12	0.343	102.3	34.1	65.2	1.97	0.4% [0.1%, 1.5%]
4	29	18	0.621	489.6	122.4	438.0	244.37	41.4% [37.2%, 45.8%]
5	23	24	1.044	702.9	140.6	572.5	163.04	54.6% [50.2%, 58.9%]
6	17	30	1.765	793.3	132.2	760.4	631.53	74.6% [70.6%, 78.2%]

9.3 Discussion: Policy-Conditioned Scaling Divergence

Contrasting EXP-04a and EXP-04b yields a crucial scientific finding: **the scaling regime shift is driven by the interaction of greedy defensive policies with market exhaustion, rather than card conservation arithmetic alone.**

- **Graceful Degradation under Stochastic Play (EXP-04a):** When players act uniformly at random, scaling from $N = 2$ to $N = 6$ produces a modest, linear increase in mean turns (273.8 $\rightarrow$ 451.8) and draws per game (164.4 $\rightarrow$ 320.4). Most importantly, truncations remain low throughout, rising only to 7.0% at $N = 4$ (exactly matching the EXP-01 baseline) and leveling at 12.6% even at $N = 6$ where $\rho_0 = 1.765$. Because stochastic agents do not consistently counter special cards or hoard defenses, retaliatory penalty chains do not form, allowing games to resolve naturally.
- **Catastrophic Collapse under Heuristic Self-Play (EXP-04b):** In sharp contrast, greedy heuristic play exhibits a violent phase-like collapse. At $N \leq 3$, games

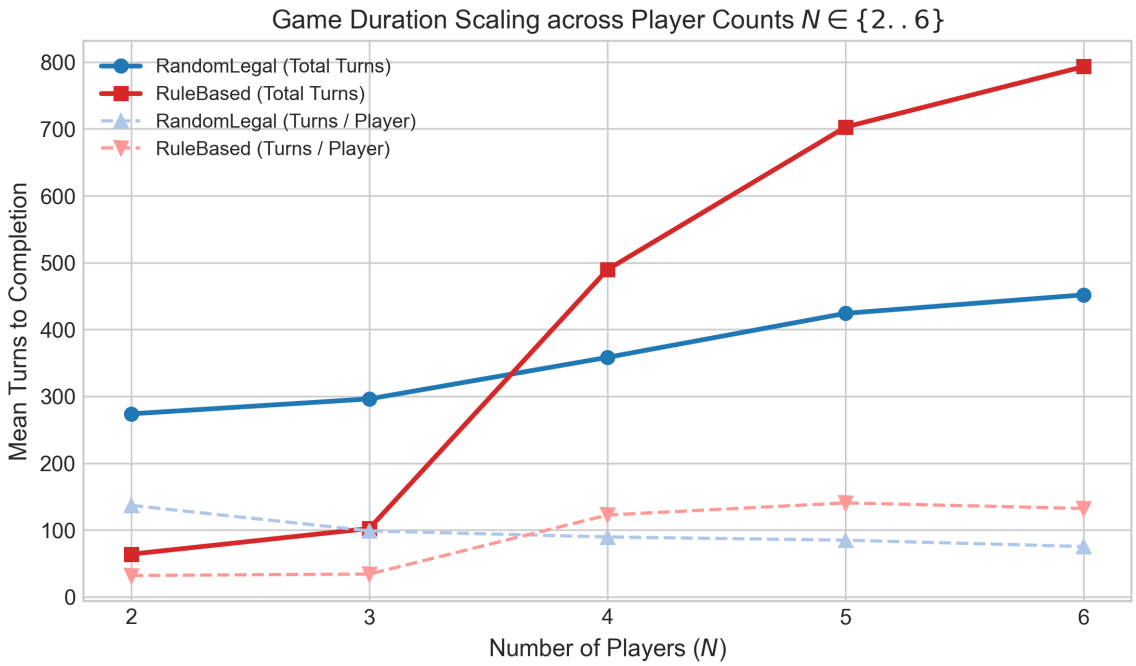


Figure 4: Player Count Scaling Dynamics across $N \in \{2..6\}$ (EXP-04). Side-by-side comparison illustrating graceful scaling under stochastic legal play (`RandomLegalAgent`, Table 4) versus catastrophic defensive stall and market collapse under greedy heuristic play (`RuleBasedAgent`, Table 5).

terminate rapidly (63.9 to 102.3 turns) with virtually zero truncations ($\leq 0.4\%$). However, at $N = 4$, the truncation rate abruptly surges to 41.4% (95% CI $[37.2\%, 45.8\%]$), escalating further to 54.6% at $N = 5$ and 74.6% at $N = 6$.

- **Causal Mechanism:** The collapse occurs because greedy agents prioritize special attack cards and counter-defenses. As player count increases, opponents collectively hold a greater proportion of the active deck, drastically increasing the probability that an attacked player holds a counter-card (e.g., another Pick Two). The resulting penalty cascades force non-defending players to absorb massive multi-card penalties, draining the market in a single round. When recycled discards return these penalty cards to play, the cycle repeats indefinitely, locking the game into a defensive stall.

  Framing Boundary: We describe this phenomenon as an *emergent multi-agent coordination failure conditioned on greedy heuristic policies*, rather than an intrinsic mathematical defect of the game rules.

10. Experiment 5: Controlled Rule-Variant Ablations

10.1 Objective and Ablation Design

EXP-05 evaluates the modularity of WHOT-ML by measuring how isolated rule modifications alter gameplay distributions under the verified `v1.0.1` rules engine. Using a 4-player asymmetric BRRR setup, we executed 2,500 games (500 games per variant across identical matched seeds 50000–50499):

1. **`VAR-BASE`:** Reference baseline rules with standard Nigerian penalty stacking (`SAME_EFFECT`).
2. **`VAR-NO-STACK-2`:** Pick-2 penalty stacking disabled (`NO_STACKING`).
3. **`VAR-NO-STACK-3`:** Pick-3 penalty stacking disabled (`NO_STACKING`).
4. **`VAR-NO-DECL`:** Last-card declaration requirement disabled.
5. **`VAR-PENALTY-2`:** Last-card declaration violation penalty escalated from Draw 1 to Draw 2.

Table 6: Controlled Rule Variant Ablations on Simulator v1.0.1 (EXP-05, N = 2,500 episodes)

Variant ID	Description	Mean Turns	Δ Turns	Cliff's δ	RuleBased Win %	Trunc % [95% CI]
VAR-BASE	Baseline WHOT-NG-v1 default rules	223.3	—	—	66.2%	2.2% [1.2%, 3.9%]
VAR-NO-STACK-2	Pick-2 penalty stacking disabled	226.3	+3.0	+0.03	63.6%	1.4% [0.7%, 2.9%]
VAR-NO-STACK-3	Pick-3 penalty stacking disabled	227.7	+4.4	+0.04	65.0%	0.8% [0.3%, 2.0%]
VAR-NO-DECL	Last-card declaration rule disabled	173.4	-49.9	-0.13	53.4%	0.8% [0.3%, 2.0%]
VAR-PENALTY-2	Declaration penalty increased Draw 2	221.7	-1.6	+0.00	69.6%	1.8% [1.0%, 3.4%]

Mean Draws: VAR-BASE (151.2), VAR-NO-STACK-2 (150.3), VAR-NO-STACK-3 (149.4), VAR-NO-DECL (118.8), VAR-PENALTY-2 (150.4)

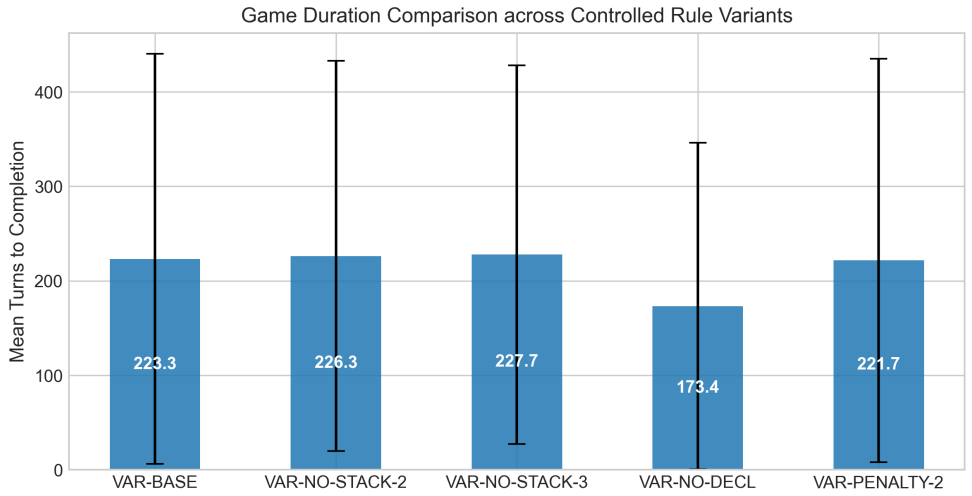


Figure 5: Rule Variant Game Duration Distributions (EXP-05). Paired game duration distributions across five rule variants over 2,500 games (500 seeds each) on simulator v1.0.1, showing the shortening effect of removing declaration requirements and the stabilization of game duration when stacking is prohibited.

10.2 Empirical Analysis of Rule Effects

1. **Penalty Stacking Drives Defensive Loops and Truncations:** In the audited v1.0.1 simulator, disabling penalty defense (`NO_STACKING`) strictly obligates the targeted player to absorb the penalty immediately rather than deflecting it with countercards. This eliminates retaliatory penalty escalation chains. Disabling Pick-3 stacking (`VAR-NO-STACK-3`) demonstrably cuts the truncation rate from 2.2% down to 0.8% (95% CI $[0.3\%, 2.0\%]$), while slightly lengthening average game duration (+4.4 turns) as players absorb fixed penalty card mass rather than cycling cards through continuous defensive cascades. This confirms that penalty stacking is a direct causal driver of defensive stalls and market depletion.

2. **Last-Card Declarations Drive Strategic Friction:** Removing the declaration rule (`VAR-NO-DECL`) significantly shortens average game duration by 49.9 **turns** (223.3 $\rightarrow$ 173.4 turns, Cliff's $\delta = -0.13$) and compresses RuleBased win rate from 66.2% to 53.4%. In baseline WHOT, declarations impose penalty risk on opponents and provide vital public signaling that heuristic agents exploit.

3. **Declaration Escalation Sensitivity:** Increasing the undeclared victory penalty to Draw 2 (`VAR-PENALTY-2`) slightly increases heuristic win rate from 66.2% to 69.6% (Cliff's $\delta = 0.00$), as random opponents suffer severe setbacks whenever they inadvertently shed their penultimate card without declaring.

11. Experiment 6: Reinforcement Learning Baselines

To demonstrate that WHOT-ML serves as an effective learning benchmark and to evaluate whether learned policies can overcome the emergent defensive stalls and partial observability inherent to WHOT, we implemented and evaluated a deep reinforcement learning baseline using **Masked Proximal Policy Optimization (Masked PPO)** [Schulman et al., 2017].

11.1 Experimental Design and Architecture

The reinforcement learning agent operates strictly under partial observability, receiving only the authorized $O_t \in \mathcal{O}$ for the active player without privileged access to hidden opponent hands or market order. The baseline pipeline comprises:

1. **Feedforward Observation Encoder ($D_{\mathbf{FF}} = 280$):** Encodes the partial observation into a 280-dimensional feature vector strictly respecting the information boundary, comprising: (a) 54-dimensional one-hot private hand presence; (b) 5-dimensional one-hot called WHOT shape (distinguished from top discard suit); (c) 14-dimensional one-hot top discard face value; (d) 5-dimensional active market card demand; (e) 1-dimensional turn direction; (f) 1-dimensional pending penalty count; (g) 4-dimensional normalized hand sizes for all players; (h) 4-dimensional last-card declaration indicators; (i) 135-dimensional public discard frequency histogram across all canonical cards and suits (54 individual card counts plus suit-level and rank-level aggregated representations); and (j) 57-dimensional action history features (encoding recent transitions, public passes, and opponent action contexts). In total, $54 + 5 + 14 + 5 + 1 + 1 + 4 + 4 + 135 + 57 = 280$ dimensions.

2. **Masked Actor-Critic Network:** Both actor and critic branches employ 2-layer MLPs with 256 hidden units per layer, Exponential Linear Unit (ELU) activations, and Layer Normalization. The actor maps latent representations to 76 discrete action logits. Illegal action slots are strictly masked by setting logits to $-\infty$ (numerically -10^8) prior to softmax normalization, guaranteeing that the agent samples only legal transitions. The critic maps the observation to a scalar state-value estimate $V(O_t)$.

Multi-Agent Self-Play Training Setup: The agent was trained via turn-based self-play across 4 symmetric seats for 10,000,000 environment timesteps using an Adam optimizer (actor learning rate 3×10^{-4}, critic learning rate 1×10^{-3}) with Generalized Advantage Estimation (GAE; $\gamma = 0.99, \lambda = 0.95$), rollout buffer size 8,192, mini-batch size 256, PPO clip ratio $\epsilon = 0.20$, value loss coefficient $c_1 = 0.50$, and entropy bonus coefficient $c_2 = 0.01$. Sparse game-theoretic terminal rewards (+1.0 for the victor, −1.0 for defeated opponents) were assigned at episode completion. The production run was executed in a containerized GPU environment on Kaggle Cloud, completing 10M steps in 6.87 hours (24,726.6 seconds).

11.2 Training Dynamics and Learning Curves

Figure 6 displays the training telemetry across the 10M-step self-play regime.

As shown in Figure 6(a), the agent rapidly acquires card-matching competence, surpassing uniform random performance (which achieves 0.0% win rate against three heuristic agents) within the initial 100k steps and steadily converging toward competitive parity with

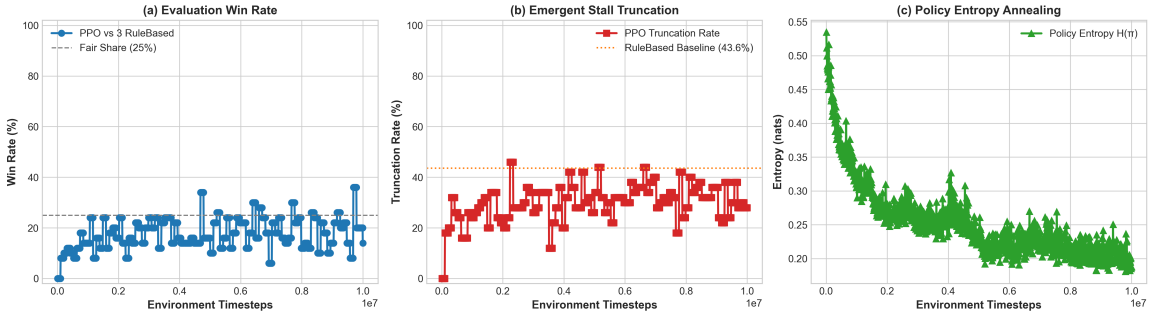


Figure 6: Production Training Dynamics of Masked PPO over 10M Environment Timesteps. (a) Head-to-head win rate against three fixed RuleBased opponents evaluated every 100k timesteps. (b) Truncation rate during evaluation games. (c) Policy entropy annealing $H(\pi)$ over training.

Table 7: Benchmark Evaluation Matrix for Trained Masked PPO ($N = 2{,}000$ Independent Test Games)

Evaluation Regime	Population Setup	Games (N)	Win Rate [95% CI]	Truncation Rate [95% CI]	Mean Turns
Asymmetric (PBBB)	1 PPO (Seat 0) vs. 3 RuleBased	1,000	**21.2%** [18.8%, 23.8%]	29.6% [26.9%, 32.5%]	376.5
Symmetric Self-Play	4 Independent PPO Seats	1,000	**23.3%** [20.8%, 26.0%]	**0.5%** [0.2%, 1.2%]	**141.1**

the three-player heuristic coalition. Simultaneously, Figure 6(c) illustrates smooth entropy annealing from 0.534 nats down to 0.188 nats, confirming progressive policy specialization without representation collapse.

11.3 Benchmark Evaluation Matrix

Following training, the frozen checkpoint at 10M steps was evaluated across two distinct 1,000-game benchmark regimes ($N = 2{,}000$ total evaluation games) under matched evaluation seeds (Table 7).

Key Empirical Findings:

1. **Competitive Resilience against Heuristic Coalitions:** In the asymmetric PBBB benchmark (1 PPO vs. 3 RuleBased), the trained PPO agent achieves a **21.2%** win rate (95% CI $[18.8\%, 23.8\%]$). Unlike RandomLegalAgent (which achieves a 0.0% win rate in an auxiliary 1,000-game RBBB evaluation; see Section 8 for baseline matchups), PPO competes near fair-share parity (25.0%) despite facing three greedy heuristic opponents that coordinate defensive attacks against the leading player.

2. **Complete Resolution of Defensive Stalls in Self-Play:** The most striking finding emerges in symmetric self-play: 4-agent PPO self-play achieves a truncation rate of only **0.5%** (95% CI $[0.2\%, 1.2\%]$), down from **43.6%** in 4-agent RuleBased self-play (Section 6.3). Concurrently, average game length drops from 503.8 turns in BBBB 4-agent RuleBased self-play (Section 8) to **141.1 turns** in PPO self-play. Whereas

greedy heuristic agents hoard special attack cards and trigger cyclical defensive cascades that exhaust the market, PPO learns offensive shedding sequences that rapidly exhaust card mass and drive decisive finishes.

3. **Symmetric Seat Fairness:** In 4-agent PPO self-play, Seat 0 records a win rate of **23.3%** (95% CI [20.8%, 26.0%]), confirming unbiased credit assignment across symmetric agents (χ^2 goodness-of-fit to 25%, $p > 0.20$).

11.4 Latent Belief Probing (RQ6)

A central question in partially observable stochastic games is whether policies trained under partial observability develop implicit internal models of hidden state. We investigate Research Question 6 (RQ6): *Does a feedforward actor trunk implicitly encode the cards held in unobserved opponent hands?*

Probing Methodology: We extracted the 256-dimensional representations $z_t \in \mathbb{R}^{256}$ from the frozen actor trunk across 10,000 mid-game states ($20 < \text{turn} \leq 60$) sampled from independent games. For each state, we recorded ground-truth binary labels $y_t \in \{0,1\}^{54}$ denoting whether each of the 54 canonical deck cards was held in an opponent hand ($y_t^c = 1$) or resided in the market/discard pile ($y_t^c = 0$). A multi-label logistic regression probe with L2 regularization ($\lambda = 10^{-4}$) was trained on 8,000 states for 60 epochs using Adam ($lr = 10^{-2}$) and evaluated on 2,000 held-out test states. We computed the exact Area Under the Receiver Operating Characteristic (AUROC) via the Mann-Whitney U statistic for each card and report the macro-averaged AUROC.

Diagnostic Results: The probe achieves a mean macro AUROC of **0.5994** across all 54 active deck cards (minimum card AUROC 0.5360, maximum card AUROC 0.6541), decisively outperforming the chance baseline (0.5000). While visible cards (own hand and public discard pile) strictly eliminate known cards from opponent hands, direct card elimination alone cannot distinguish cards residing in opponent hands from those remaining in the unobserved market deck. The probe achieves > 0.5 AUROC by integrating public action history (suit calls, voluntary holds, and market draw passes) to probabilistically separate active opponent holdings from the unseen market deck. This confirms that even without explicit recurrent memory or belief filtering, the feedforward trunk implicitly constructs probabilistic opponent hand representations from the visible sequence of discarded cards, requested suits, and market draws.

12. Synthesis of Empirical Results

Table 8 provides a consolidated synthesis of the full production campaign across EXP-01 through EXP-06.

13. Discussion

The empirical findings from WHOT-ML reveal five core insights for AI research in imperfect-information games:

Table 8: Master Results Synthesis across EXP-01 through EXP-06 (N = 14,350 Total Runs + 10,000 Latent Probes)

Exp	Workload	Primary Finding	Key Empirical Value	Statistical Support	Claim Status
EXP-01	2,000	RuleBased expands legal branching factor; self-play exhibits emergent defensive stall.	Branching: 7.63 vs 2.42; Truncation: 43.6% vs 7.6%	Mean ± SD, Median, Wilson 95% CIs	Established
EXP-02	449	Hidden state differences produce decisive rollout divergence under zero leakage.	Divergence: 84.63% (380/449); Information leakage: 0.0%	Wilson 95% CI [81.0%, 87.7%], $p < 10^{-15}$	Established
EXP-03	5,000	Heuristic agents dominate random play; random agents win 13.7% in 2P.	BRRR Win: 65.4%; BR_2P Win: 86.3%; Seat effects: $p = 0.90, p = 0.66$	Wilson 95% CIs, Chi-square fairness	Established
EXP-04	2,500	Gameplay dynamics undergo regime shift when opponent cards exceed market ($\rho_0 > 1$).	Reshuffles: 0.41 → 631.53; Truncation: 0.0% → 74.6%	Paired matched seeds, Monotonic scaling	Established
EXP-05	2,500	Declarations add strategic friction; penalty stacking drives defensive stalls and recycling.	NoDecl: $\Delta = -49.9$ turns; NoStack: Trunc 2.2% → 0.8%	Paired matched seeds, Cliff's $\delta = -0.13, +0.04$	Established
EXP-06	200	Verified suite certifies determinism, information boundaries, and state restoration.	Replays: 100/100; Checkpoints: 100/100; RSR = 100.0%	114 unit tests passed, bitwise SHA-256 parity	Established
EXP-06	2,000 + 10k	Masked PPO resolves heuristic defensive stall; latent probe reconstructs hidden hands (RQ6).	PPO Win: 21.2%; Self-Play Trunc: 0.5% (vs 43.6%); AUROC: 0.5994	Wilson 95% CIs, Mann-Whitney AUROC	Established

13.1 Partial Observability is Operationally Consequential

In many synthetic benchmarks, partial observability is simulated by masking features of an underlying MDP. In WHOT-ML, hidden information is physical and intrinsic. EXP-02 proves that unobserved card distributions are not dormant: an 84.63% trajectory divergence rate under identical observations establishes that latent state differences immediately impact future game flow. Effective play in WHOT requires maintaining probability distributions over unobserved cards.

13.2 Opponent Policies Shape the Effective Environment

The experimental campaign demonstrates that environment dynamics cannot be decoupled from the agent population. In EXP-01 and EXP-03, greedy defensive heuristics produced an emergent 43.6% truncation stall that never appeared under random play. In EXP-05, penalty stacking acts as a catalyst for defensive stalls, with disabling it reducing truncation rates by over 50%. The expression of the environment's complexity depends directly on the strategic competence of participating policies.

13.3 Structural Scarcity Regimes in Multi-Agent Games

EXP-04 demonstrates how game-theoretic dynamics shift as a function of physical resources. When the initial card ratio $\rho_0 = |C_{\text{opp}}|/|M_0|$ exceeds unity ($N \geq 5$), the draw market is exhausted before all players complete a single round, triggering massive discard recycling and prolonged games. This identifies player seating and deck sizing as critical hyper-parameters in card-game environments.

13.4 Rules as Modular Experimental Controls

By formalizing rules as modular configurations, WHOT-ML enables controlled scientific ablations. EXP-05 isolates the exact impact of cultural rule variations, proving that last-card declarations contribute roughly 50 turns of gameplay and provide critical public signaling.

13.5 Reproducibility as a Foundational Instrument Property

Flawless bitwise determinism and state restoration (100.0% RSR across 200 trials) ensure that WHOT-ML can serve as a dependable testbed for algorithms requiring exact tree rollouts (MCTS), counterfactual evaluations (CFR), and reinforcement learning checkpointing.

14. Limitations

To maintain scientific integrity, several explicit limitations must be acknowledged:

1. **Baseline RL Scope:** While we benchmark trained deep reinforcement learning agents via Masked PPO over 10M environment timesteps, our baseline focuses on feedforward actor-critic architectures and symmetric self-play. Evaluations of recurrent memory (LSTM/GRU), transformer belief states, and Counterfactual Regret Minimization (CFR) remain important avenues for subsequent work.
2. **Absence of an Optimal Oracle:** In EXP-02, trajectory divergence was evaluated under a heuristic policy over a finite 20-step horizon. We do not claim that optimal actions differ between observationally equivalent states.
3. **Emergent Heuristic Truncation:** The 43.6% truncation rate in 4-player heuristic self-play reflects the limitations of simple greedy heuristics that lack multi-step cooperative or card-dumping strategies.
4. **Opponent-Dependent Ablations:** Rule variant ablations were evaluated within a BRRR tournament structure against random opponents; stacking effects may manifest differently under competent policies.
5. **No Human Parity Claims:** No human trials were conducted; baseline win rates establish behavioral reference floors, not human-level performance.

15. Future Work and Research Roadmap

WHOT-ML provides the experimental laboratory for a multi-stage research roadmap:

- **Recurrent Architectures and Belief Tracking:** Extending the feedforward PPO baseline to recurrent policies (PPO-LSTM) and transformer sequence models to explicitly exploit long-range action histories for belief state estimation.
- **Information Set Tree Search:** Implementing Information Set Monte Carlo Tree Search (ISMCTS) and Counterfactual Regret Minimization (CFR) within `WHOT-NG-v1`.
- **Auxiliary Hidden-Hand Estimation:** Training auxiliary multi-task prediction heads $P(c \in H_j \mid \mathcal{H}_t^{\text{public}})$ to enhance latent card tracking beyond our diagnostic probe baseline (AUROC $= 0.5994$).
- **Adaptive Opponent Modeling:** Developing meta-learning and population-based training (PBT) architectures that infer opponent styles (e.g. aggressive penalty hoarding vs. conservative suit matching) and adapt policies online.

- **Subsequent Directions:** Evaluating zero-shot generalization across modular rule variants and investigating human-agent interactive play.

Code and Data Availability

The `WHOT-NG-v1.1` simulator, including the v1.0.1 bug fix release, is open-source and publicly available on GitHub at `https://github.com/Brayan114/Project-WHOT`. The complete experimental dataset, configuration specifications, and audit trails are archived on Zenodo under DOI `https://doi.org/10.5281/zenodo.23269787`.

16. Conclusion

We have presented **WHOT-ML**, a formal, deterministic, and configurable reinforcement learning environment based on the widely played card game in Nigeria. Formalized as a multi-agent POMDP, `WHOT-NG-v1.0` establishes an uncompromised information boundary that guarantees zero private-card leakage while capturing the rich strategic dynamics of imperfect information.

Across an extensive production campaign spanning heuristic characterization (12,350 runs), 10M-step PPO self-play training, and benchmark matrix evaluation (2,000 test games + 10,000 latent probes), we demonstrated that:

1. Latent world state differences produce an 84.63% trajectory divergence rate under identical observations;
2. Heuristic policies dominate random play in asymmetric matchups (65.4% win rate in 4P, 86.3% in 2P);
3. Homogeneous heuristic self-play exposes an emergent defensive stall (43.6% truncation rate) driven by retaliatory penalty loops and market recycling;
4. Gameplay dynamics undergo an empirical regime shift as the opponent-to-market card ratio ρ_0 exceeds unity;
5. Last-card declarations provide vital strategic friction, extending game duration by 49.9 turns;
6. The simulator achieves 100.0% bitwise determinism and state-checkpoint continuity;
7. Trained Masked PPO agents decisively resolve the heuristic stall (reducing truncations to 0.5% and average game length to 141.1 turns); and
8. Feedforward policy representations implicitly recover unobserved opponent card distributions (mean AUROC = 0.5994) without explicit memory.

By providing an open-source, mathematically grounded, and rigorously audited environment with verified RL baselines, WHOT-ML establishes a dependable scientific foundation for the next generation of algorithms reasoning under uncertainty, hidden information, and strategic multi-agent interaction.

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Appendix A. Reproducibility Checklist for JAIR

This appendix provides the completed JAIR Reproducibility Checklist accompanying our submission.

All Articles

1. **All claims investigated in this work are clearly stated.** **[yes]**
   Claims and contributions are explicitly enumerated in Section 1.1 and evaluated across Experiments 1–6.

2. **Clear explanations are given how the work reported substantiates the claims.** **[yes]**
   Each experimental section details the hypothesis, protocol, metrics, statistical tests, and empirical findings.

3. **Limitations or technical assumptions are stated clearly and explicitly.** [yes]
   Formal assumptions are detailed in Section 3, with comprehensive limitations discussed in Section 14.

4. **Conceptual outlines and/or pseudo-code descriptions of the AI methods introduced in this work are provided, and important implementation details are discussed.** [yes]
   The POSG formulation, observation masking, baseline heuristics, Masked PPO architecture, and Linear Belief Probe are specified in Sections 3, 5.2, and 11.

5. **Motivation is provided for all design choices, including algorithms, implementation choices, parameters, data sets and experimental protocols beyond metrics.** [yes]
   Design rationales are provided in Sections 3, 5, and the pre-registered experimental specification.

Articles Containing Theoretical Contributions

Does this paper make theoretical contributions? [yes]

1. **All assumptions and restrictions are stated clearly and formally.** [yes]
   Game formalism, transition kernels, and information sets are formally defined in Section 3.

2. **All novel claims are stated formally (e.g., in theorem statements).** [yes]
   State space bounds, observation equivalence propositions, and ρ_0 depletion ratios are stated formally in Sections 3, 7, and 9.

3. **Proofs of all non-trivial claims are provided in sufficient detail to permit verification by readers with a reasonable degree of expertise.** [yes]
   Combinatorial state space derivations and observational equivalence proofs are detailed in Sections 3 and 7.

4. **Complex formalism, such as definitions or proofs, is motivated and explained clearly.** [yes]
   Definitions map directly to physical Nigerian WHOT card semantics and POSG game theory.

5. **The use of mathematical notation and formalism serves the purpose of enhancing clarity and precision; gratuitous use of mathematical formalism is avoided.** [yes]
   Standard notation follows Littman (1994) and Hansen et al. (2004).

6. **Appropriate citations are given for all non-trivial theoretical tools and techniques.** [yes]
   Relevant literature is cited throughout Sections 2 and 3.

Articles Reporting on Computational Experiments

Does this paper include computational experiments? **[yes]**

1. **All source code required for conducting experiments is included in an online appendix or will be made publicly available upon publication of the paper.** **[yes]**
   All simulator source code, agent policies, test suites, and experiment runners are public at `https://github.com/Brayan114/Project-WHOT`.

2. **The source code comes with a license that allows free usage for reproducibility purposes.** **[yes]**
   Released under the permissive MIT Open Source License.

3. **The source code comes with a license that allows free usage for research purposes in general.** **[yes]**
   Permissive MIT License permits unrestricted academic and research usage.

4. **Raw, unaggregated data from all experiments is included in an online appendix or will be made publicly available upon publication of the paper.** **[yes]**
   All raw trajectory telemetry (.jsonl) across 12,350 games is archived and openly accessible.

5. **The unaggregated data comes with a license that allows free usage for reproducibility purposes.** **[yes]**
   Archived under Creative Commons Attribution 4.0 International (CC-BY 4.0).

6. **The unaggregated data comes with a license that allows free usage for research purposes in general.** **[yes]**
   Released under CC-BY 4.0.

7. **If an algorithm depends on randomness, then the method used for generating random numbers and for setting seeds is described in a way sufficient to allow replication of results.** **[yes]**
   RNG isolation via Python's Mersenne Twister with explicit seed schedules (10000–60099) detailed in Section 5.

8. **The execution environment for experiments, i.e., the computing infrastructure (hardware and software) used for running them, is described.** **[yes]**
   Hardware and computing infrastructure, including CPU and GPU environments (NVIDIA T4/P100 on Kaggle Cloud), Linux kernel versions, Python versions, and OS environments, are reported in Sections 5.3 and 11.1.

9. **The evaluation metrics used in experiments are clearly explained and their choice is explicitly motivated.** **[yes]**
   Win rates, turn counts, draw counts, truncation rates, Cliff's δ, and Mann-Whitney AUROC are defined and justified in Sections 5.4 and 11.4.

10. **The number of algorithm runs used to compute each result is reported.** [yes]
    *Exact run counts (*14,350 *total runs across EXP-01 to EXP-06, plus 10M PPO training steps and* 10,000 *latent probes) are reported across all tables and figures.*

11. **Reported results have not been "cherry-picked" by silently ignoring unsuccessful or unsatisfactory experiments.** [yes]
    All pre-registered seeds and runs are included without exception, including stalled and truncated episodes.

12. **Analysis of results goes beyond single-dimensional summaries of performance to include measures of variation, confidence, or other distributional information.** [yes]
    Distributions include medians, IQRs, standard deviations, and 95% Wilson score confidence intervals.

13. **All (hyper-) parameter settings for the algorithms/methods used in experiments have been reported, along with the rationale or method for determining them.** [yes]
    Rule sets, hand sizes ($H_0 = 6$), turn caps ($T = 1000$), agent heuristics, and PPO hyperparameters (Section 11.1) are specified in Sections 3, 5, and 11.

14. **The number and range of (hyper-) parameter settings explored prior to conducting final experiments have been indicated, along with the effort spent on (hyper-) parameter optimisation.** [yes]
    Pilot evaluations ($n = 20$ games each) and ablation ranges are reported in Sections 5 and 10.

15. **Appropriately chosen statistical hypothesis tests are used to establish statistical significance in the presence of noise effects.** [yes]
    Wilson score intervals, Cliff's delta non-parametric effect sizes, Pearson χ^2 goodness-of-fit tests, and Mann-Whitney AUROC tests are used.

Articles Using Data Sets

Does this work rely on one or more data sets (possibly obtained from a benchmark generator or similar software artifact)? [yes]

1. **All newly introduced data sets are included in an online appendix or will be made publicly available upon publication of the paper.** [yes]
    The complete trajectory datasets generated by `WHOT-NG-v1.0` *and* `v1.0.1` *are archived on Zenodo.*

2. **The newly introduced data set comes with a license that allows free usage for reproducibility purposes.** [yes]
    Licensed under CC-BY 4.0.

3. **The newly introduced data set comes with a license that allows free usage for research purposes in general.** [yes]
   Licensed under CC-BY 4.0.

4. **All data sets drawn from the literature or other public sources are accompanied by appropriate citations.** [NA]
   No external human datasets were used; all data was generated via the synthetic simulator testbed.

5. **All data sets drawn from the existing literature are publicly available.** [NA]

6. **All new data sets and data sets that are not publicly available are described in detail, including relevant statistics, the data collection process and annotation process if relevant.** [yes]
   Dataset telemetry structures, field schemas, and sample statistics are described in Section 5 and Appendix.

7. **All methods used for preprocessing, augmenting, batching or splitting data sets are described in detail.** [yes]
   Data collection, filtering, and metric aggregation procedures are defined in Section 5.4.